Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT (DS) Sem-I Remedial April-June 2024

U71A1IP1 INTRODUCTION TO PROGRAMMING-I

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any Five)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Explain the structure of C program.	06	CO2	BTL2
2	What is an algorithm? Write down its characteristics and rules for developing an algorithm.	06	CO1	BTL3
3	What is a variable? How one can declare a variable in C language? Discuss the rules to declare a variable in C.	06	CO2	BTL3
4	Describe the features of C language.	06	CO2	BTL4
5	Explain different increment and decrement operators in C with example.	06	CO3	BTL2
6	What is the use of ternary operator? Describe its usage and demonstrate its implementation.	06	CO4	BTL1, BTL3
7	What is the use of loop? Explain for loop with suitable example.	06	CO5	BTL1, BTL3
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Compare compiler and interpreter.	06	CO2	BTL4
2	Develop an algorithm and draw a flowchart to read two numbers from the user and perform arithmetic operations on those numbers	06	CO1	BTL6
3	Write an algorithm and draw a flowchart to swap two numbers.	06	CO1	BTL6

Q.3	**Answer the following: (Any Two)**	**12**		
1	What is the use of operator? Explain different arithmetic and logical operators used in C language.	06	CO3	BTL1, BTL2
2	What is the use of printf() and scanf() functions? Also discuss different format specifiers.	06	CO3	BTL1, BTL2
3	What is the use of switch statement? Demonstrate its implementation.	06	CO4	BTL3
Q.4	**Answer the following: (Any One)**	**06**		
1	Compare while and do…while loop.	06	CO5	BTL4
2	Explain the use of break and continue statement with example.	06	CO5	BTL2, BTL3

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc (Data Science) Sem-I Remedial April-June 2024

U71A2DA1 DATA ANALYSIS - I

Time: 3 Hours

Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Difference between MS Word & Excel.	06	CO1, CO2	BTL5
2	Explain Mail Merge steps with example.	06	CO1	BTL2
3	Describe a short note on Power point layout.	06	CO4	BTL4
4	Difference between Word & PowerPoint.	06	CO3	BTL5
5	Explain Date-Time Function with example.	06	CO2	BTL6
6	Describe Summarizing Worksheets with Hidden and Filtered Rows.	06	CO3	BTL4
7	Elaborate Database tools for Microsoft Excel and Microsoft Access.	06	CO5	BTL5
Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Define Terms:- Header, Footer, Endnote, Footnote, Hyperlink.	06	CO1	BTL3
2	How to Protect Documents in Microsoft office? Write down the steps for password protection.	06	CO1	BTL3
3	What is Addressing modes? Explain types of addressing modes.	06	CO2	BTL1
Q.3	**Answer the following: (Any Two)**	**12**		
1	Describe Details:- Naming Range, Pivot Table, Goal seek	06	CO3	BTL3
2	Write down steps for Conditional formatting in Excel.	06	CO3	BTL3
3	How to insert Chart and Table in Microsoft word.	06	CO4	BTL6
Q.4	**Answer the following: (Any One)**	**06**		
1	What is MS Access? Benefits of using Microsoft Access.	06	CO5	BTL5
2	How to import and export data in Microsoft access	06	CO5	BTL4

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT(DS) Sem-I Remedial April-June 2024

U71A3WD1: WEB DESIGNING-I

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I				
Q-1	**Answer the following. (Any 5)**	**Marks 30**	**CO**	**Bloom's Taxonomy**
1	Differentiate between the Internet, Intranet, and Extranet.	06	CO1	BTL5
2	Illustrate the process of creating a table in HTML, including specifying rows, columns, and cell contents.	06	CO2	BTL6
3	Justify the significance of HTML5 in addressing the limitations of previous HTML versions.	06	CO3	BTL4
4	Differentiate between different methods of applying CSS styles to HTML elements.	06	CO4	BTL5
5	Why is JavaScript considered advantageous for client-side scripting compared to other languages?	06	CO5	BTL3
6	Illustrate the process of setting background styles using CSS, demonstrating different techniques such as color, image, and gradient backgrounds.	06	CO4	BTL5
7	Classify web pages into static and dynamic types. Explain the differences between them.	06	CO1	BTL4

Section – II				
Q-2	**Answer the following (Any Two)**	**Marks 12**	**CO**	**Bloom's Taxonomy**
1	Differentiate between block and inline elements in HTML. Give examples of each.	06	CO2	BTL5
2	Elaborate on the concept of web design principles and their importance in creating user-friendly websites.	06	CO1	BTL4
3	Explain in detail how HTML forms are created and processed, including the use of form elements and attributes.	06	CO2	BTL6
Q.3	**Answer the following: (Any Two)**	**12**		
1	Discuss the benefits and challenges associated with implementing HTML5 drag and drop features in web applications.	06	CO3	BTL3
2	Discuss the impact of CSS on web design standards and best practices, including its role in responsive web design.	06	CO4	BTL4
3	Elaborate on the features and benefits of using web storage and web workers in HTML5 applications.	06	CO3	BTL4
Q.4	**Answer the following: (Any One)**	**06**		
1	Justify the use of cookies in JavaScript and their importance in maintaining state and user preferences across sessions.	06	CO5	BTL3
2	Illustrate the usage of regular expressions in JavaScript.	06	CO5	BTL4

**** End of the Paper ****

Bloom's Taxonomy BTL5

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. IT(DS) - I (CBCS) Remedial April - June 2024

U71A4IDS: INTRODUCTION TO DATA SCIENCE

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

Q-1	Answer the following. (Any 5)	Marks 30	CO	Bloom's Taxonomy Pattern
1	How is Data Science applied in various fields and what challenges may arise in each?	06	CO1	BTL1
2	What are the different roles and responsibilities of Data Analyst?	06	CO1	BTL2
3	Analyze and elaborate on the data cleaning process, including techniques and best practices.	06	CO2	BTL4
4	List the features of the Python programming language and provide brief descriptions for each.	06	CO3	BTL1, BTL2
5	Define regression and explain its importance and common use cases.	06	CO4	BTL1, BTL2
6	What are the key principles behind ridge and lasso regression techniques and how they address multicollinearity and overfitting issues in linear regression model.	06	CO4	BTL5
7	What is overfitting and underfitting, and how can they impact the effectiveness of machine learning models?	06	CO5	BTL1

Section – II

Q-2	Answer the following (Any Two)	Marks 12	CO	Bloom's Taxonomy Pattern
1	Describe the steps involved in the Data Science process.	06	CO1	BTL2
2	Analyze the key components of Data Science, explaining their relationship in solving complex problems.	06	CO1	BTL4
3	Explain data preprocessing steps.	06	CO2	BTL5

Q.3	Answer the following: (Any Two)	12		
1	Discuss types of EDA techniques and their respective applications in gaining a deeper understanding of data?	06	CO3	BTL3
2	Explain the significance of box plots, pivot tables, heat maps, histograms, and line graphs in EDA	06	CO3	BTL3
3	What is the difference between simple and multiple linear regression, and how are they each applied in analyzing relationships between variables?	06	CO4	BTL5

Q.4	Answer the following: (Any One)	06		
1	What is the purpose of cross-validation, and how does it contribute to model selection and evaluation?	06	CO5	BTL2
2	Discuss different evaluation metrics used for assessing the performance of machine learning models, such as accuracy, precision, recall, and F1-score, and explain when each metric is most appropriate	06	CO5	BTL2, BTL4

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT DS – Semester -I

Remedial – April-June 2024

U71A5FM: FUNDAMENTAL OF MATHEMATICS

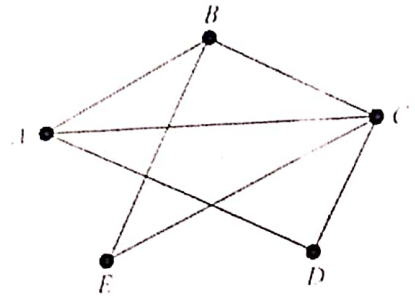
Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2-Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

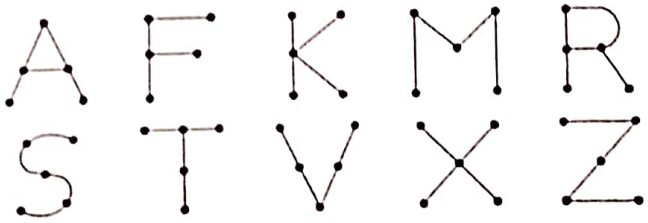
Section – I

Q-1	Answer the following. (Any 5)	Marks 30	CO	Bloom's Taxonomy Pattern
1	Let $U = \{1,2,3,4,5,6,7,8\}$ be the Universal set and let $A = \{1,3,5,6\}$, $B = \{2,4,6,7\}$, $C = \{2,3,7,8\}$ Find (a) $B \cup C$ (b) $A \cap C$ (c) A/C (d) B/A (e) $B \oplus C$	06	CO1	BTL1, BTL2, BTL6
2	Each student in Liberal Arts at some college has a mathematics requirement A and a science requirement B. A poll of 150 sophomore students shows that: 70 completed A, 50 completed B, 21 completed both A and B. Use a Venn diagram to find the number of students who have completed: (a) At least one of A and B; (b) exactly one of A or B; (c) neither A nor B.	06	CO2	BTL4 & BTL5 & BTL6
3	Define square matrix.Also prove that A.(B+C)=A.B+A.C where $A=\begin{bmatrix}1 & 2 & 2\\ 2 & 2 & 3\\ 1 & -2 & 2\end{bmatrix} \& B=\begin{bmatrix}1 & 2 & 3\\ 3 & 1 & 1\\ 2 & 2 & 3\end{bmatrix} \& C=\begin{bmatrix}1 & 0 & 2\\ 2 & 4 & 0\\ -1 & 2 & 3\end{bmatrix}$	06	CO3	BTL1, BTL2, BTL6
4	Define the following function with example. (a) Injective function (B)Surjective function (c) In to function (d) Identity function (e) Equal function (f) Constant function	06	CO2	BTL2 &BTL3

5	Define Inverse Function. Also (a) If f:N $\rightarrow$N and f(x)=x+1. Does f^{-1} exist ? why? (b) If f:R$\rightarrow$R and f(x)=2x-3. Then find $f^{-1}(x)$ & $f^{-1}(9)$	06	CO4	BTL 3&
6	Consider the graph G  (a)Describe G formally, that is, find the set V (G) of vertices of G and the set E(G) of edges of G. (b) Find the degree of each vertex and verify the sum of degree is twice the number of edges.	06	CO3	B B B
7	Find the number n of distinct permutations that can be formed from all the letters of each word: (a) LEADER; (b) UNUSUAL; (c) RUMOUR.	06	CO2	BT BT BT

Section – II

Q-2	Answer the following (Any Two)	Marks 12	CO	Bloom's Taxonomy Pattern
1	Define Cartesian Product of two sets. If $A = \{1,2,4,6\}$, $B = \{1,3,5,7\}$ Find $(i) A \times B$ $(ii) B \times A$ $(iii)(A \times B) \cap (B \times A)$	06	CO1	BTL1, BTL2, BTL6
2	In a survey of 120 people, it was found that: 65 read Newsweek magazine, 46 read Time, 41 read Fortune ,27 read both Newsweek and Time, 25 read both Newsweek and Fortune, 15 read both Time and Fortune, 8 read all three magazines. (a) Find the number of people who read at least one of the three magazines. (b) Fill in the correct number of people in each of the eight regions of the Venn diagram where N, T , and F denote the set of people who read Newsweek, Time, and Fortune, respectively. (c) Find the number of people who read exactly one magazine.	06	CO4	BTL2 &BTL3 &BTL5
3	Show that A.B ≠ B.A. For the matrix $A = \begin{bmatrix} 3 & 2 & -1 & -2 \\ 4 & 3 & 2 & 6 \\ -1 & 5 & 7 & 3 \\ 2 & 5 & 3 & 4 \end{bmatrix}$ & $B = \begin{bmatrix} -3 & -3 & -3 & -3 \\ 4 & 2 & 5 & 6 \\ -2 & 5 & 6 & -4 \\ -5 & 5 & -5 & 5 \end{bmatrix}$	06	CO1	BTL1, BTL2, BTL6

Q.3	Answer the following: (Any Two)	12		
1	Define: domain. Also If $f: R \to R, g: R \to R$ & $h: R \to R$ is define as $f(x) = x^2 + x, g(x) = x^2 - 1$ & $h(x) = 4x^2$. Find fog,foh,fo(goh),gog and hoh.	06	CO3	BTL4 & BTL5 & BTL6
2	Define one-one and Onto function. Also if $f: R \to R, g: R \to R$ & $h: R \to R$ be a functions define as f(x)=4x/3, g(x)=3x and h(x)=4x-2 then prove that (fog)oh =fo(goh).	06	CO2	BTL1 & BTL2
3	 Which of the above graphs are isomorphic?	06	CO4	BTL1 & BTL2

Q.4	Answer the following: (Any One)	06		
1	Prove that: $\binom{n+1}{r} = \binom{n}{r-1} + \binom{n}{r}$.	06	CO4	BTL3,BTL4
2	A class contains 11 students with 6 men and 5 women. Find the number n of ways to: (a) Select a 4-member committee from the students. (b) Select a 4-member committee with 2 men and 2 women. (c) Elect a president, vice president, and treasurer.	06	CO4	BTL3, BTL4, BTL6

**** End of the Paper ****

Enrollment No._____

GANPAT UNIVERSITY

B.Sc.IT (Data Science) Sem-I Remedial Theory Exam April-June 2024

U71B6CS1: LANGUAGE & COMMUNICATION SKILLS-I

Time: 3 Hours | Total: 60 Marks

Instructions:

1. Figures to the right indicate full marks, Course Outcomes and BL Pattern Level.
2. This Question paper has two sections. Each section should be written in a separate answer book.
3. Section-I consist of 07 questions of 6 marks each with options respectively. Section-II consist of 03 questions, Question 2 of 12 Marks, Question 3 of 12 Marks and Question 4 of 6 Marks with options respectively.
4. Be precise and to the point in your answer.
5. CO is Course Outcome from the Subject Syllabus.
6. Bloom's Taxonomy Based Assessment Pattern: BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create.

Section – I

Q-1	Answer the following. (Any 5)	Marks 30	CO	Bloom's Taxonomy
1	Discuss the use of subordinating conjunctions in complex sentences.	06	CO1	BTL1,4
2	Discuss the characteristics and advantages of verbal and non-verbal communication.	06	CO2	BTL2,6
3	Describe strategies for overcoming language barriers in multicultural environments.	06	CO3	BTL2,5
4	Describe strategies for improving reading comprehension, considering factors such as skimming, scanning, and inference.	06	CO4	BTL2,3
5	Discuss the characteristics of an erudite speaker, emphasizing qualities such as clarity, confidence, and coherence.	06	CO5	BTL1,2
6	Differentiate between the structures of formal and informal writing. Explain the equation **E=A+I** in terms effective writing.	06	CO4	BTL2,6
7	Explain the concepts of coherence and cohesion in writing and their significance in conveying ideas clearly.	06	CO2	BTL1,3

Section – II

Q-2	Answer the following (Any Two)	Marks 12	CO	Bloom's Taxonomy
1	Choose the correct option for the following questions:	06	CO1	BTL1,3

i. Choose the sentence with the correct article usage:
a) I saw an elephant at zoo.
b) I saw elephant at zoo.
c) I saw the elephant at zoo.
d) I saw elephant at the zoo.

ii. Select the sentence used with correct verb tense:
a) They have visited Paris last summer.
b) They visit Paris last summer.
c) They visited Paris last summer.
d) They visiting Paris last summer.

iii. He is not only smart, ____ also hardworking.
a) but b) and c) so d) or

iv. Choose the correct collocation:
a) She made a noise. b) She did a noise.
c) She had a noise. d) She took a noise.

v. Identify the sentence with the correct word order:
a) The dog chased the cat around the yard.
b) Chased the cat around the yard the dog.
c) Around the yard, the dog chased the cat.
d) The cat chased the dog around the yard.

vi. We should ____ the meeting for later.
a) put off b) put on
c) put up d) put away

2	Do as directed:	06	CO1	BTL1,3

i. Fill in the blanks with 'a', 'an', or 'the' as necessary:
She wants to become ___ engineer.

i. Correct the verb tense in the sentence:
We has gone to the market.

ii. Rewrite the sentence using a suitable conjunction:
Kaya will go for a walk. It stops raining.

iii. Correct the collocation usage in the sentence:
She made a decision.

iv. Re-arrange the words to form a meaningful sentence:
dog / chased / cat / the / the / garden / in the/ went/ to/ park/ children.

v. Choose the correct phrasal verb:
The meeting ___ (put off/called off) due to bad weather.

3	Discuss the key elements of communication and their roles in conveying messages.	06	CO2	BTL2,4

Q.3	**Answer the following: (Any Two)**	**12**		
1	Define listening skill and describe the process of active listening. Explain the rudiments of effective listening, and how do they contribute to comprehension.	06	CO3	BTL2,4
2	Define reading skill and discuss its purposes and types. How do different reading techniques contribute to effective comprehension?	06	CO4	BTL2,6
3	Discuss the significance of possessing effective interpersonal oral conversation skills.	06	CO4	BTL2,6
Q.4	**Answer the following: (Any One)**	**06**		
1	Explain the importance of effective writing skills in conveying messages clearly and persuasively.	06	CO5	BTL1,2
2	Write a job application letter to a software development company, expressing interest in the software designer position advertised. Detail your qualifications, skills, and enthusiasm for the role.	06	CO5	BTL2,4

**** End of the Paper ****